8. A town has a total population of 40,000 out of which 400 people own cars, 10,000 people own bicycles and 300 people own both cars and bicycles. How many in the town do not own either?

9. A survey was taken of 200 males and 200 females to determine preferences amongst four different brands of soap, A, B, C and D. The responses to the question, "which soap do you like best?" are tabulated below:

	A	B	C	D
Male	75	45	30	50
Female	60	30	40	70

(*i*) How many males are in the set $E = A \cup B$? Verbally describe this set.
(*ii*) How many males are in the set $F = (A \cup B) \cup C$? Verbally describe this set.
(*iii*) Sets E and F are as defined in (i) and (ii). Verbally describe $(E \cup F')$. How many are in this set?

10. Consider the situation in Example 1.5.1. Write the statement "All cigarette smokers are either pipe smokers or wine drinkers" in set notation.

11. Define the sets S, P and C as follows:

S = Set of surgeons
P = Set of physicians
C = Set of cardiologists.

"Some physicians are cardiologists and not all cardiologists are surgeons. Hence some physicians are surgeons." Verify the truth of this statement using set theory.

12. Let T = Set of people who use train service
B = Set of people who use bus service
P = Set of people who play cards
S = Set of people who smoke
R = Set of people who read mystery novels.

Translate the following into set theoretic symbols:
(*i*) All people who use train service and play cards.
(*ii*) The persons who use bus services and read mystery novels but do not smoke.
(*iii*) The persons who do not smoke but use both bus and train services.
(*iv*) All those who play cards but do not smoke.
(*v*) Those who do not use train service but read mystery novels and smoke during their bus journey.

13. A statistician conducted a survey of 500 people and found that 300 liked Taj shaving soap and 270 liked Taj blades. He concluded that at least 70 people liked both Taj blades and Taj soap. Do you agree with him? Justify your answer.

14. Krishna Iron and Steel Company produces iron rods of length 5 feet, radius 2 inches and weight 400 kg. After the production, the quality controller measures the above to check if they conform to standards. On a certain day there were 14 rods classified as defective by the quality controller. The causes of defect were classified as follows:

Defects	No. of rods
Weight	7
Radius	5
Length	8

Also there were 2 rods with defects in weight and length, 1 in length and radius; and 1 in weight and radius; and 1 with all the three defects. Draw the Venn diagram. How many rods were there with defect in:
(*a*) weight alone?
(*b*) radius alone?
(*c*) length alone?

15. A survey was conducted in a city to study the preference for three newspapers: Hindustan Herald (H), Bharat News (B) and India Tribune (T). The total number of persons interviewed were 200. 92 people said that they read Hindustan Herald, 86 read Bharat News, and 83 read India Tribune. Also 25 of them read Hindustan Herald and Bharat News, 27 read Hindustan Herald and India Tribune, 26 read Bharat News and India Tribune and 9 read all the three papers. How many elements are there in each of the following sets:

(*i*) H only
(*ii*) B only
(*iii*) $H \cup B$
(*iv*) $H \cap B$
(*v*) $T \cap (H \cup B)$
(*vi*) Readers of exactly two newspapers
(*vii*) Readers of exactly one newspaper
(*viii*) Readers of no newspaper.

16. In a competitive examination which consisted of three tests, viz. general knowledge, arithmetic and English, the number of participants were 3,000. Only 2,270 participants were able to get through the arithmetic test, 750 through both arithmetic and general knowledge, 450 through both general knowledge and English, 400 through all the three tests and 1,000 through arithmetic and English. There were 250 participants who got through general knowledge alone but not through the other two and 200 passed in English only.

(*i*) How many were in a position to get through general knowledge.
(*ii*) How many participants failed in all the three subjects?
(*iii*) How many got through only arithmetic?

17. A company manufactures soap, powder and toothpaste and wishes to know about its market in a particular area. A survey was conducted covering a population of 1,899 persons. From the survey report the manager came to know that 30 persons had been using all the 3 products, 757 persons had been using soap, 574 persons powder, 132 persons soap and powder, 155 persons soap and toothpaste, 472 persons toothpaste and 47 persons toothpaste and powder. Also it was stated that 398 persons did not use any of their products. After having gone through these statements, the manager concluded the statements to be wrong. Do you agree with the manager?

18. In a survey of 940 factory workers, 680 owned cycles, 380 owned both cycles and radios, 120 workers owned cycles and houses but not radios, 620 owned radios and 70 owned all the three and 250 owned houses and 110 owned houses and radios.

(*i*) How many workers owned cycles alone?
(*ii*) How many workers owned houses alone?
(*iii*) How many workers did not own any one of these three things?

19. A company desires to send a few of their executives for training programmes and it has stipulated that the executives belonging to *all* the following categories are eligible to be considered for the training programme.

Category	Description
A	Graduates
B	Between 30-35 years in age
C	Have at least 5 years' experience with the company
D	Have good record of work in the company

The company has in all 250 executives. There are 200, 150, 180 and 120 executives in the categories A, B, C and D respectively. The number of executives in various combinations of categories are shown below:

Combination of categories	No. of executives
A and B	80
A and C	101
A and D	94
B and C	74
B and D	87
C and D	104
A, B and C	52
A, B and D	67
B, C and D	49
A, C and D	25
None of A, B, C and D	12

How many executives are eligible to be considered?

20. If $A \subset B$ and $C \subset D$, then show that $(A \times C) \subset (B \times D)$

21. If $A = \{1, 2, 3\}$ and $B = \{a, b\}$

(*a*) List the elements of the Cartesian product $A \times B$.

(*b*) List the elements of $A \times A$.

22. Examine whether the following sets are finite or infinite:

(*i*) The set of points on the circumference of a circle of radius two centimeters.

(*ii*) The set of lines parallel to a given line.

(*iii*) The set of words in this book.

(*iv*) The set of real numbers between zero and one.

(*v*) The set of integers.

(*vi*) The set of citizens of India.

23. Define the set $H = \{1, 2, 3, 4, 5, 6, 7\}$

(*a*) Is $H_1 = \{1, 7, 2\}$, $H_2 = \{4, 5\}$, $H_3 = \{3, 6\}$ a partition of H?

(*b*) Is $H_1 = \{1, 7, 2\}$, $H_2 = \{4, 5\}$, $H_3 = \{3, 6, 7\}$ a partition of H?

24. Let the set $H = \{1, 2, 3, 4\}$ consist of four elements.

(*a*) How many different partitions of H can be formed?

(*b*) List out the partitions.

25. Consider a Venn diagram formed by 5 sets. How many disjoint subsets of the universal set S can be identified in the diagram?